Water Demand

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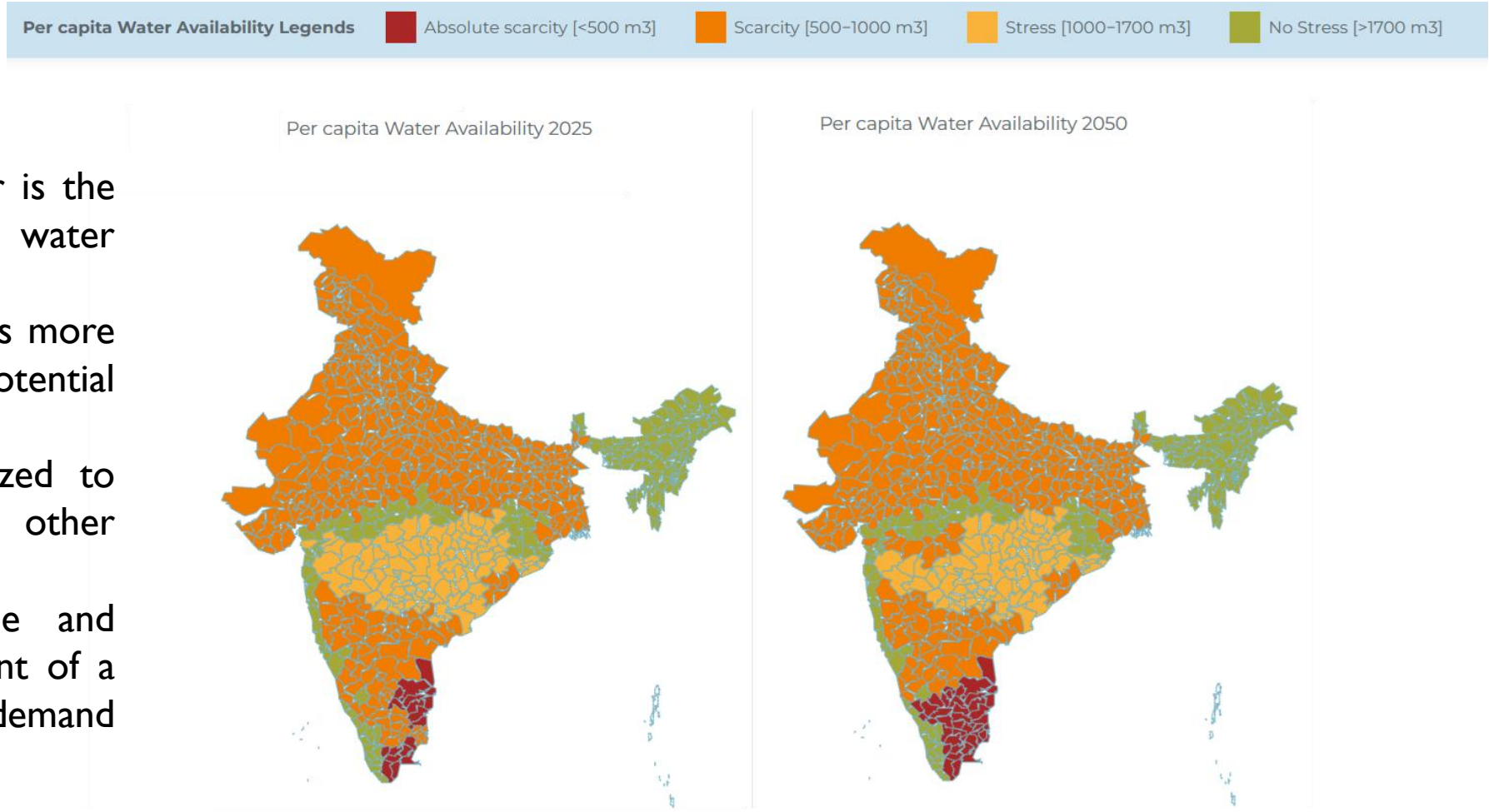
Water Demand

- Estimation of demand for water is the key parameter in planning a water supply scheme.
- The agriculture sector consumes more than 80 percent of total water potential created in our country.
- The remaining portion is utilized to meet domestic, industrial and other demands.
- The improvement in life-style and associated industrial development of a nation push up the per capita demand for water.



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Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977), 2. Per capita water availability 2025 & 2050, https://iced.niti.gov.in/climate-and-environment/water/per-capita-water-availability, Accessed on- 9 February, 2025

Various Types of Water Demand

1. DOMESTIC WATER DEMAND

- The total domestic water consumption usually amounts to 50 to 60% of the total water consumption
- The I.S. Code lays down a limit of water consumption between 135 to 225 litre per capita per day (lpcd).
- Under ordinary conditions (as per I.S. code) the minimum domestic water demand for a town or a **city with full flushing system should be taken at 200 lpcd.**
- Although it can be reduced to **135 lpcd for economically weaker sections and LIG colonies** (Low Income Group) depending upon prevailing conditions.

Table showing Minimum domestic water Consumption (Annual Average) for Indian Towns and Cities with Full Flushing Systems as per IS 1172-1993

Use	Consumptions (lpcd)
Drinking	5
Cooking	5
Bathing	75
Washing of clothes	25
Washing of utensils	15
Washing and cleaning of houses and residences	15
Lawn watering and gardening	15
Flushing of water closets, etc.,	45
Total	**200**

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Various Types of Water Demand

1. DOMESTIC WATER DEMAND

- Although it can be reduced to **135 lpcd for economically weaker sections and LIG colonies** (Low Income Group) depending upon prevailing conditions.

Table showing Minimum domestic water Consumption (Annual Average) for Weaker Section and LIG Colonies in Small Indian Towns and Cities

Use	Consumptions (lpcd)
Drinking	5
Cooking	5
Bathing	55
Washing of clothes	20
Washing of utensils	10
Washing and cleaning of houses and residences	10
Flushing of water closets, etc.,	30
Total	**135**

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Various Types of Water Demand

2. INDUSTRIAL WATER DEMAND

- The industrial water demand represents the water demand of industries which are either existing or likely to be started in future, in the city for which water supply is being planned.

- This quantity varies with the number and types of industries present in the city. This consumption, under ordinary conditions is 50 lpcd.

Water Required by Certain Important Industries

Name of Industry	Unit of Production	Approximate Quantity of Water required per unit of production/raw material in kilo litres
1. Automobiles	Vehicle	40
2. Fertilizers	Tonne	80 - 200
3. Leather	Tonne (or 1000 kg)	40 (or 4)
4. Paper	Tonne	200 - 400
5. Petroleum Refinery	Tonne (Crude)	1 - 2
6. Sugar	Tonne (Crushed cane)	1 - 2
7. Textile	Tonne (goods)	80 - 140
8. Distillery (Alcohol)	kilo litre	122 - 170

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

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3. INSTITUITIONAL AND COMMERCIAL WATER DEMAND

- The water requirements of institutions such as hospitals, hotels, restaurants, schools and colleges, railway station etc. should also be assessed and provided for in addition to domestic and industrial water demands discussed above.

- On an average, a per capita demand of 20 lpcd is usually considered to be enough to meet such commercial and institutional water requirements.

- Although this demand may be as high as 50 lpcd for highly commercialized cities.

Water demand of certain commercial establishments:

(i)	Offices	45 *l*pcd
(ii)	Schools	45 to 135 *l*pcd
(iii)	Hostels	135 *l*pcd
(iv)	Hotels	180 *l*pcd
(v)	Hospitals	450 *l*pcd
(vi)	Cinema halls	15 *l*pcd

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Various Types of Water Demand

4. DEMAND FOR PUBLIC USES

- This includes water requirement for parks, gardening, washing of roads etc. On this account a nominal amount not exceeding 5% of the total consumption may be provided.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

Various Types of Water Demand

5. FIRE DEMAND

- The quantity of water required for extinguishing fire is not very large, the total amount of water consumption for a city of 50 lakh population hardly **amounts to 1 lpcd**. But this water should be easily available and kept always stored in storage reservoirs,

- Three jet streams are simultaneously thrown from each hydrant; one on the burning property, and one each on adjacent property on either sides of the burning property. The discharge of each stream should be about 1100 litres/minute.

- The minimum water pressure available at fire hydrants should be of the order of 1 to 1.5 kg/cm^2 and should be maintained even after 4 to 5 hours of constant use of fire hydrant.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Various Types of Water Demand

5. FIRE DEMAND

- Rate of fire demand is worked out by following formulas:

1. **Kuichling's Formula:**

$$Q = 3182\sqrt{P}$$

Where, Q = Amount of water required in litres/minute ; P = Population in thousands.

2. **Freeman Formula:**

$$Q = 1136\left[\frac{P}{5} + 10\right]$$

3. **National Board Formula:**

(A) For a central congested high valued city:

(i) When population ≤ 2 lakhs: $Q = 4637\sqrt{P}\left[1 - 0.01\sqrt{P}\right]$

(ii) When population ≥ 2 lakhs: A provision for 54600 litres per minute may be made with an additional provision of 9100 to 36400 litres/minute for a second fire.

(B) For a residential city:

(i) For small or low buildings: Q = 2200 litres/minute.

(ii) For large or higher buildings: Q = 4500 litres/minute.

4. **Buston's Formula:**

$$Q = 5663\sqrt{P}$$

P and Q have the same meaning as above.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

Numerical

Compute the 'fire demand' for a city of 2 lakh population by any two formulae (including that of the National Board of Fire Underwriters).

(i) The rate of fire demand as per National Board of Fire Underwriters Formula for a central congested city whose population is less than or equal to 2 Lakh is given by

$$Q = 4637\sqrt{P}\left[1-0.01\sqrt{P}\right] = 4637\sqrt{200}\left[1-0.01\sqrt{200}\right]$$

$$= 56303.08\ l/\text{min} = \frac{56303.08 \times 60 \times 24}{10^6}\text{MLD} = \mathbf{81.08\ MLD}$$

(ii) Kuchling's formula, $Q = 3182\sqrt{P} = 3182\sqrt{200}\ l/\text{min}$; $R = 45000.27\ l/\text{m} = \mathbf{64.8\ MLD}$

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)
2. IS 1172-1993, Code of basic requirements for water supply, drainage and sanitation

IMPORTANT FACTS

Dual flush system

- The modern-day double flush toilets come with two different types of levers or buttons. One is larger, while the other is smaller and each button is connected to its own exit valve.
- The larger lever is to flush out around 6 to 9 litres of water, whereas the smaller lever is to flush out around 3 to 4.5 litres of water. Clearly, the larger one is to flush solid waste and the smaller one is to flush liquid waste.

Source: 1. National Programme on Climate Change and Human Health, National Centre for Disease Control, Government of India, Guidelines for Green and Climate Resilient Healthcare Facilities, https://ncdc.mohfw.gov.in/wp-content/uploads/2024/06/Guidelines-for-Green-and-Climate-Resilient-Healthcare-Facilities_Feb2023.pdf (2023).

Various Types of Water Demand

6. WATER DEMAND FOR LOSSES & THEFT

- This may be as high as 15% of total demand.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

POPULATION FORECASTING

1. Arithmetic Increase Method

This method assumes that the population increases at a constant rate.

$$\frac{dp}{dt} = \text{Constant}$$

Forecasted population (P_n) after 'n' decades from the last known census is given by

$$P_n = P_o + n\bar{x}$$

where, P_0 = Population at last known census.

$\bar{x}$ = Average (Arithmetic mean) of population increase in the known decades.

n = Number of decades between last census and future.

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

Numerical

The population of 5 decades from 1930 to 1970 are given in the table. Find out the population after one, two and three decades beyond the last known decade, by using arithmetic increase method.

Year	Population
1930	25,000
1940	28,000
1950	34,000
1960	42,000
1970	47,000

The future populations are now computed by using equation as

$P_n = P_0 + n \cdot \bar{x}$

∴ (a) Population after 1 decade beyond 1970

$= P_{1980} = P_1 = P_{1970} + 1 \cdot \bar{x}$

$= 47{,}000 + 1 \times 5500 = 52{,}500$

(b) Population after 2 decades beyond 1970

$= P_{1990} = P_2 = P_{1970} + 2 \cdot \bar{x}$

$= 47{,}000 + 2 \times 5500 = 58{,}000$

(c) Population after 3 decades beyond 1970

$= P_{2000} = P_3 = P_{1970} + 3 \cdot \bar{x}$

$= 47{,}000 + 3 \times 5500 =$ **63,500**

Year (1)	Population (2)	Increase in population (x) (3)
1930	25,000	
1940	28,000	3000
1950	34,000	6000
1960	42,000	8000
1970	47,000	5000
Total		22,000
Average increase per decade ($\bar{x}$)		$\bar{x} = \frac{22000}{4} = 5{,}500$

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

POPULATION FORECASTING

2. Geometric Increase Method

- This is compounding rate method also called *'uniform increase method'*.
- The forecasted population (P_n) after n decades is given by

$$P_n = P_0\left(1+\frac{r}{100}\right)^n$$

Where, P_0 = Population at last known census ; r = Assumed growth rate (%)

r can be calculated by following methods.

$$r = \frac{\text{Increase in population}}{\text{Original population}} \times 100 \text{ for each decade}$$

Knowing $r_1, r_2, r_3 \ldots r_n$ for each decade the average value of r can be found by

Geometric average method: $r = (r_1 \times r_2 \times r_3 \times \ldots \times r_n)^{\frac{1}{n}}$

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

Numerical

Determine the future population of a satellite town by the Geometric increase method for the year 2011, given the following data-

Year	Population in Thousand
1951	93
1961	111
1971	132
1981	161

Constant growth rate, assumed for future

r = geometric mean of past growth rates = $\sqrt[3]{19.35 \times 18.92 \times 21.97}$ = 20.03 % per decade

The population after n decades in now given by equation

$$P_n = P_0\left(1+\frac{r}{100}\right)^n$$

P_{2011} = Population after 3 decades from 1981

$$= P_{1981}\left(1+\frac{20.03}{100}\right)^3 = 1{,}61{,}000(1.2003)^3 = \mathbf{2{,}78{,}417}$$

Table

Year	Population in thousand	Increase in Population in thousand	%age increase in population = growth rate $= \frac{col(3)}{col(2)} \times 100$
1951	93		
		18	
1961	111		19.35
		21	
1971	132		18.92
		29	
1981	161		21.97

Source: 1. Santosh Kumar Garg, Water supply engineering, Khanna Publishers (1977)

Thank you

For any questions, please feel free to reach out to me.